

Predictive Model for CO₂ Corrosion Engineering in Wet Natural Gas Pipelines[☆]

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ABSTRACT

Starting from a "worst case" corrosion rate prediction with the deWaard-Milliams equation, correction factors can be applied to quantify the influence of environmental parameters and corrosion product scale formed under various conditions. Equations are proposed for each factor. A low-temperature scale formed in condensing water can cause a reduction in the top-of-the-line corrosion rate in pipelines. At higher temperatures, a more protective scale forms even under high liquid flow rates. The decrease of corrosion rates caused by dissolved Fe is accounted for with a pH correction factor. The effect of the presence of a liquid hydrocarbon phase is included. Equations are presented that enable the effect of glycol injection on corrosion to be calculated along the length of a pipeline. Combinations of effects in the model are discussed.

KEY WORDS: carbon dioxide, wet gas, condensation, iron carbonate, glycol, corrosion model

INTRODUCTION

The feasibility of transporting wet, untreated natural gas is becoming an important factor for the development of gas producing fields. The prediction of the corrosivity associated with the presence of even small amounts of CO₂ can often play a decisive role in the determination of this feasibility. To this end, systematic review is needed of every parameter that

can affect the corrosion rate of carbon steel and that should be included in a predictive model. For many of these parameters, their effects are not known with a high degree of certainty. In these cases, the model should give conservative answers, that is, pessimistic corrosion rates, by using equations that account for the available data in a conservative manner.

BASIC CORROSION RATE EQUATION AND ITS MODIFICATION

As a starting point for the prediction of corrosion rates of carbon steel in CO₂-containing environments, the deWaard-Milliams equation and the corresponding nomogram¹ has gained wide acceptance. This equation contained two temperature-dependent terms. A re-evaluation of the data used has shown that it is possible to simplify the equation for the "nomogram corrosion rate" V_{nomo} to:

$$\log V_{\text{nomo}} = 5.8 - \frac{1710}{T} + 0.67 \log (p\text{CO}_2) \quad (1)$$

where T = temperature, °K, and $p\text{CO}_2$ (= mol% CO₂ x total pressure P) is the partial pressure of CO₂, in bar. The nomogram itself can be simplified with straight lines only instead of using a curved temperature scale and is given in Figure 1. The resulting corrosion rates, which do not differ significantly from those obtained with the old equation, represent a "worst case" prediction.

The proposed model systematically modifies this prediction by multiplying V_{nomo} with factors, each of which is associated with the effect of one physical or

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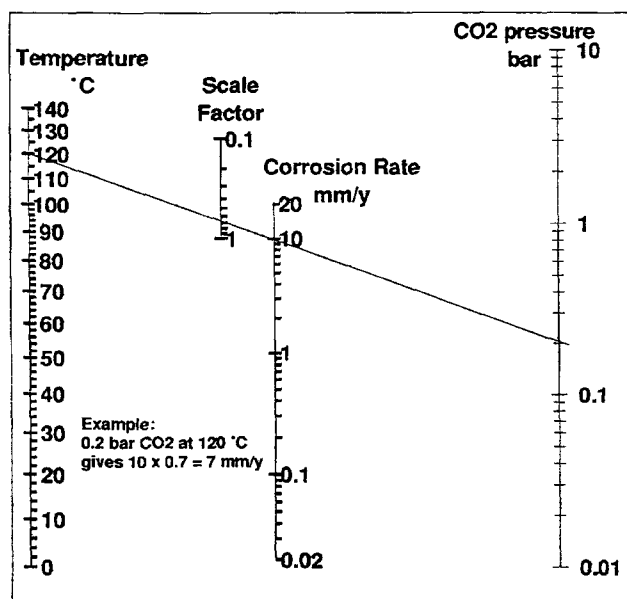


FIGURE 1. Nomogram for CO₂ corrosion.

chemical effect that can cause deviations from Equation (1). These factors are in almost all cases less than one and will tend to reduce the corrosion rates predicted with this equation, which in many cases, would otherwise give over-conservative results. Only for the case of correction for pH of a medium that is under-saturated with corrosion product, FeCO₃ or Fe₃O₄, a correction factor >1 may be obtained.

EFFECT OF TOTAL PRESSURE

An increase in total pressure P of the gas will lead to an increase in corrosion rate, because $p\text{CO}_2$ in Equation (1) will increase in proportion. However, with increasing pressure, the nonideality of the natural gas will play an increasing role, and instead of the CO₂ partial pressure, the CO₂ fugacity $f\text{CO}_2$ should be used:

$$\log V_{\text{nomo}} = 5.8 - \frac{1710}{T} + 0.67 \log(f\text{CO}_2) \quad (1a)$$

$$f\text{CO}_2 = a \times p\text{CO}_2 \quad (2)$$

where a = fugacity coefficient, analogous to activity coefficients in solutions. The fugacity coefficient can be calculated by solving the equation of state for the mixture of CO₂ and natural gas. For the binary system CO₂-CH₄ calculations were made using an approach described by Lammers.² The results given in Figure 2 can be used as a conservative estimate for a ; the presence of other gases will generally further reduce the fugacity coefficient. As a first estimate the effect on corrosion rate can be

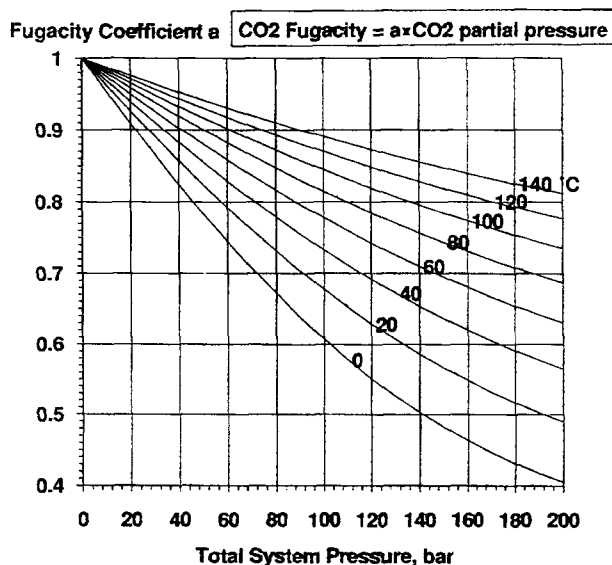


FIGURE 2. Fugacity coefficient for CO₂ in methane for gas mixtures with less than 5 mol% CO₂.

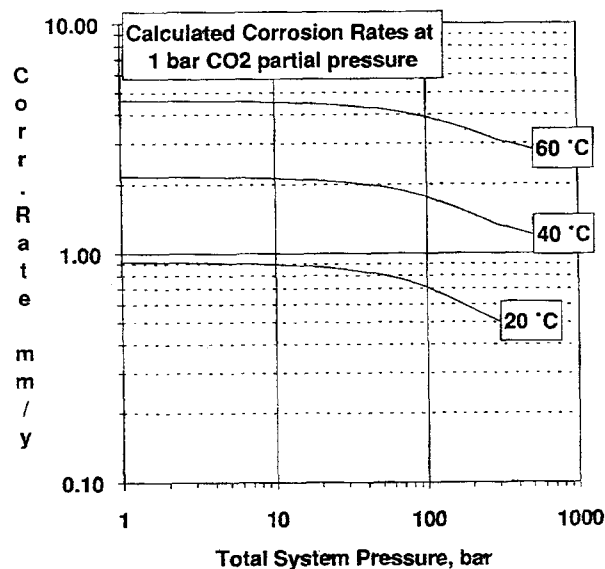


FIGURE 3. Examples of effect of total pressure on corrosion rate at a constant CO₂ partial pressure of 1 bar.

described by the multiplier F_{system}

$$\log F_{\text{system}} = 0.67 \left(0.0031 - \frac{1.4}{T} \right) P \quad (3)$$

to be applied to V_{nomo} in Equation (1). The effect on corrosion is illustrated in Figure 3, where corrosion rate is given for a constant CO₂ partial pressure, but with increasing total pressure. No experimental evidence is available for this effect. It is of such fundamental nature, however, that it should be taken into account in the model.

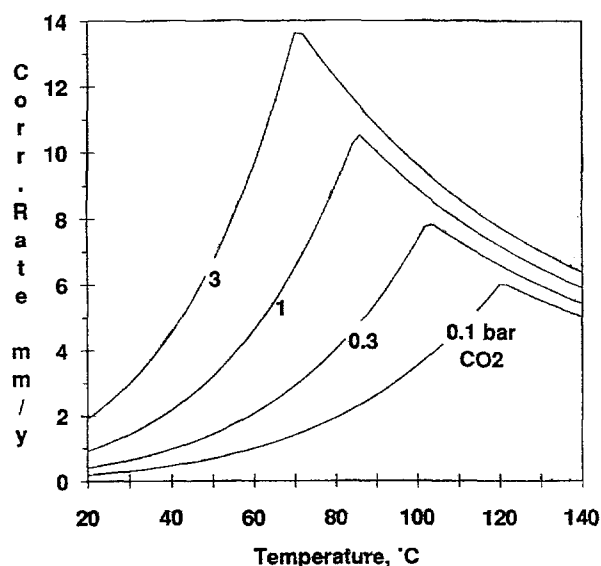


FIGURE 4. Effect of high-temperature corrosion product film calculated with scale factor (Equation [4]).

HIGH-TEMPERATURE PROTECTIVE FILMS

The effect of the formation of protective films has been studied extensively.³⁻⁶ The precipitation of FeCO_3 (or Fe_3O_4) in itself does not necessarily result in the formation of a protective film. At lower temperatures (e.g., less than 60°C) the corrosion product film has a smudge-like appearance and is easily removed by flowing liquids. At higher temperatures, the film is different in texture, is more protective, and is less easily washed away. Further increase in temperature results in lower corrosion rates and the corrosion rate goes through a maximum. This temperature is referred to here as the scaling temperature. At this temperature, the local pH and Fe^{++} concentration formed at the steel's surface are such that a protective film is formed. It is likely to be dependent on flow rate; a higher flow rate will result in a higher scaling temperature. Also, a higher bulk pH will tend to lower this temperature. Data on CO_2 corrosion rates for high temperatures and CO_2 pressures have been reported by Ikeda et al.⁶ Since they were obtained at high flow rates while covering the widest published ranges in terms of temperatures and CO_2 pressures, these data have been used as the basis for the equation for the scale factor F_{scale} in the model. This was done by calculating $\log(V_{\text{obs}}/V_{\text{nomo}})$ (V_{obs} = observed corrosion rate, V_{nomo} from Equation [1a]) at temperatures above 60°C and finding a best fit to $1/T$ and $\log(f\text{CO}_2)$ by multidimensional regression analysis with a computer spreadsheet. With a small shift to obtain a conservative envelope for Ikeda's data, the formula for the scale factor used in the model becomes

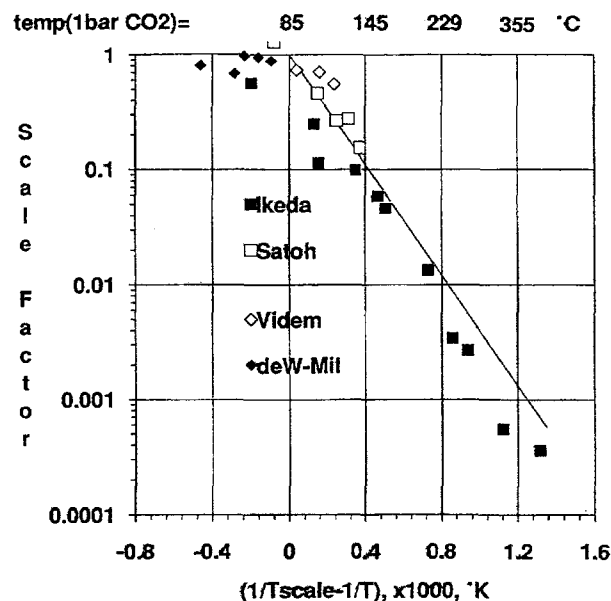


FIGURE 5. Comparison of proposed equation for the scale factor with some published data.^{1,4,6,7}

$$\log F_{\text{scale}} = \frac{2400}{T} - 0.6 \log(f\text{CO}_2) - 6.7 \quad (4)$$

with a maximum value for F_{scale} of 1. F_{scale} is set = 1 when Equation (4) would give a higher value. This factor is used to correct the corrosion rate given by Equation (1a) by multiplying V_{nomo} with F_{scale} . The temperature at which $\log(F_{\text{scale}}) = 0$ is the scaling temperature

$$T_{\text{scale}} (^{\circ}\text{K}) = \frac{2400}{6.7 + 0.6 \log(f\text{CO}_2)} \quad (5)$$

This is the temperature where the corrosion rate goes through a maximum. In the present approach the maximum will appear as a sharp peak (Figure 4). The scaling temperature decreases with increasing CO_2 pressure.

For temperature/pressure combinations below the scaling temperature, the nomogram corrosion rate does not need to be corrected for this effect. Equations (4) and (5) can be combined to

$$\log(F_{\text{scale}}) = 2400 \left(\frac{1}{T} - \frac{1}{T_{\text{scale}}} \right) \quad (6)$$

where $T > T_{\text{scale}}$, otherwise $F_{\text{scale}} = 1$.

Figure 5 shows the satisfactory fit of this equation with the data from Ikeda and others.^{1,4,6,7} Videm's results from once-through tests were corrected for lower pH associated with the absence of Fe^{++} (Equation [9]); they also fit predictions rather well, even though these tests were done at very high flow rates up to 20 m/s,

while most other tests were at 2 to 4 m/s. The nomogram in Figure 1 contains the scale factor: when the scale factor line is intersected, the factor is to be applied, otherwise no correction needs to be made.

At temperatures exceeding the scaling temperature, corrosion rates tend to decrease to close to zero with time. The reliability of a complete protection afforded by scale cannot be sufficiently quantified at present, and the scale factor gives a minimum estimate for its protectiveness. It should be noted that when brines such as formation water are likely to be present, the model will suggest $F_{\text{scale}} = 1$ because of the risk of breakdown of the film.⁸

The high-temperature scale can be eroded by high-speed liquids. Practical velocities for smooth flow in systems with liquid only are often too low to achieve this; only the impact of high-speed liquid droplets can be expected to damage the scale. Assuming the occurrence of disturbed flow in practical systems, the suggestion made by Smart⁹ is followed that the onset of erosion corrosion is coincident with the transition to the annular mist flow regime in multiphase flow. With the superficial liquid velocities associated with wet gas transport, this transition can be expected for superficial gas velocities between 15 and 20 m/s. Above 20 m/s, it is prudent to set

$$F_{\text{scale}} = 1.$$

RELATION BETWEEN Fe^{++} CONCENTRATION AND pH

Depending on temperature and CO_2 pressure, corrosion of steel in the CO_2 /water system can lead to the

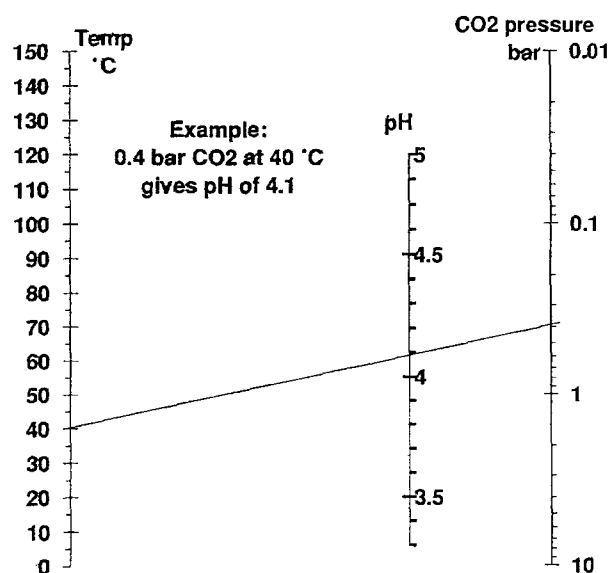


FIGURE 6. Nomogram for pH of water and CO_2 as function of CO_2 pressure (fugacity) and temperature.

formation of both FeCO_3 and of magnetite, Fe_3O_4 (and under some conditions, even the formation of Fe_2O_3 may be possible).

During experiments with a constant volume of water (at constant CO_2 pressure), the Fe^{++} concentration will increase while the H^+ concentration will decrease with time until the solution is saturated with FeCO_3 or Fe_3O_4 . This precipitation does not necessarily result in the formation of a protective film, and corrosion can continue in such a solution. When saturation with FeCO_3 occurs, further addition of iron will not change the Fe^{++} concentration or the pH in the solution any further. In the case of saturation with Fe_3O_4 , however, the pH and the Fe^{++} concentration can keep increasing in most cases, although the rate of change will be lower than without saturation.

Initially, the pH is that of water and CO_2 only¹

$$\text{pH}(\text{water}+\text{CO}_2 \text{ only}) = 3.71 + 0.00417 t - 0.5 \log(f\text{CO}_2) \quad (7)$$

This equation is also given in the form of a nomogram in Figure 6.

The corrosion rate is often observed to first increase with time and then to pass through a broad maximum or plateau. Hausler et al.¹⁰ also reported the existence of a steady-state corrosion rate, which is difficult to maintain without pH control. At this point in the test, the activation of the surface is complete, but the pH has shifted away from the original value for CO_2 and water. More importantly, it is to be expected that the increase of Fe^{++} in the solution will also have contributed to the change of the corrosion rate. It is postulated that the occurrence of a plateau or steady-state corrosion

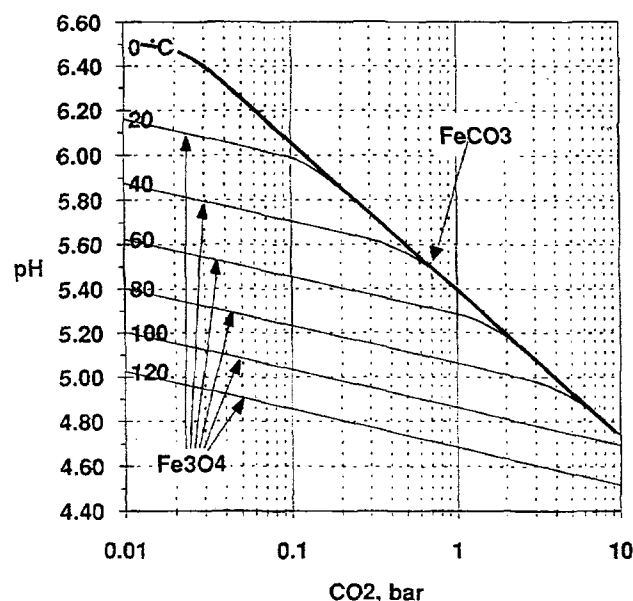


FIGURE 7. Calculated pH of water at saturation with FeCO_3 or Fe_3O_4 .

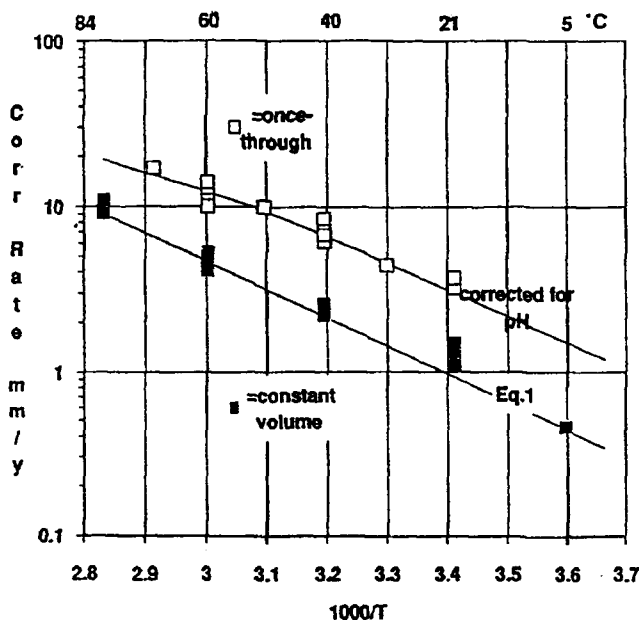


FIGURE 8. Comparison of "once-through" and "constant volume" corrosion tests as function of reciprocal temperature. All tests at 1 bar CO_2 . Lines drawn are for Equation (1) with and without pH correction factor of Equation (10).

rate coincides with the onset of saturation with FeCO_3 or Fe_3O_4 . The data on which Equation (1) was originally based are steady-state corrosion rate readings during constant volume tests with a Fe^{++} concentration in the test solution reaching the values needed for $\text{FeCO}_3/\text{Fe}_3\text{O}_4$ saturation, while the pH increased to a value denoted as pH_{sat} .

Experimentally, the pH shift when this plateau is reached ranges from 0.5 to 1.6. This agrees with thermodynamic calculations based on literature data¹¹ for the pH at the onset of FeCO_3 or Fe_3O_4 saturation for different temperatures, pH_{sat} . The results (for 10% NaCl) can be approximated by

$$\begin{aligned} \text{smallest value of} \\ \text{pH}_{\text{sat}} &= 1.36 + \frac{1307}{t+273} - 0.17 \log(f\text{CO}_2) \\ \text{and} \\ \text{pH}_{\text{sat}} &= 5.4 - 0.66 \log(f\text{CO}_2) \end{aligned} \quad (8)$$

which is shown in Figure 7. In this equation, the first formula refers to the formation of Fe_3O_4 , the second to FeCO_3 ; the smallest pH_{sat} refers to corrosion product that is more stable and more likely to form first. These results confirm earlier calculations published by Dunlop et al.¹² showing that FeCO_3 is only stable in a rather narrow window of low temperatures and high CO_2 pressures with Fe_3O_4 being formed otherwise.

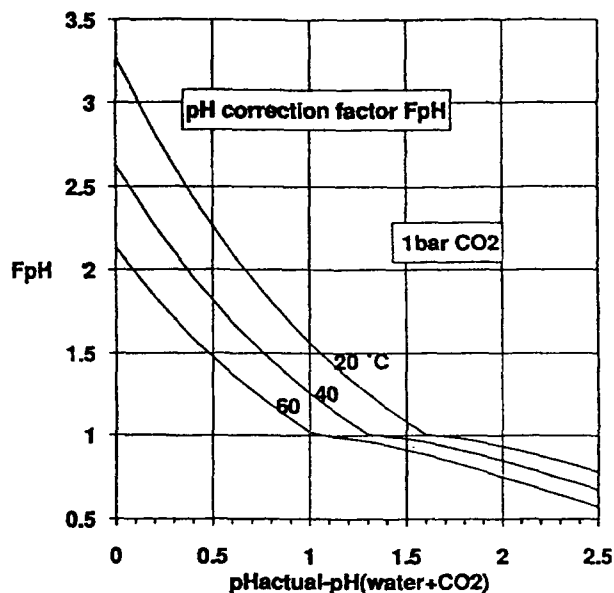


FIGURE 9. Nomogram correction factor for pH shift w.r.t. pH of water and CO_2 only.

EFFECT OF Fe^{++} AND pH ON CORROSION RATE

The contamination of the CO_2 solution with corrosion product reduces the corrosion rate. Without the presence of corrosion products, much higher corrosion rates are possible, as is demonstrated by comparison of data from "once-through" and from constant volume tests in Figure 8. In order to describe this effect, the pH shift caused by the presence of dissolved Fe^{++} (as predicted from Equations [7] and [8]) was chosen as a parameter. The following correlation was found

$$\log F_{\text{pH}} = 0.32 (\text{pH}_{\text{sat}} - \text{pH}_{\text{act}}) \quad (9)$$

$$(\text{pH}_{\text{sat}} > \text{pH}_{\text{act}})$$

where F_{pH} is a correction factor for the nomogram corrosion rate of Equation (1), and pH_{act} is the actual pH. The correction factor resulting from Equation (9) is shown in Figure 9. When $\text{pH}_{\text{act}} = \text{pH}_{\text{sat}}$, no correction is needed, that is, $F_{\text{pH}} = 1$. When pH_{act} is the initial pH, the corrosion rates in "once-through" tests were a factor 2.2 to 3.3 higher for a temperature range of 80 to 20°C. When $\text{pH}_{\text{act}} > \text{pH}_{\text{sat}}$ because of the presence of alkaline substances, NaHCO_3 , for example, the validity of Equation (9) is doubtful since it could relate to over-saturation w.r.t. Fe^{++} . In this case the formula proposed by Dugstad and Videm¹³ can be used for solutions that are Fe^{++} saturated:

$$\log F_{\text{pH}} = -0.13 (\text{pH}_{\text{act}} - \text{pH}_{\text{sat}})^{1.6} \quad (10)$$

$$(\text{pH}_{\text{sat}} < \text{pH}_{\text{act}})$$

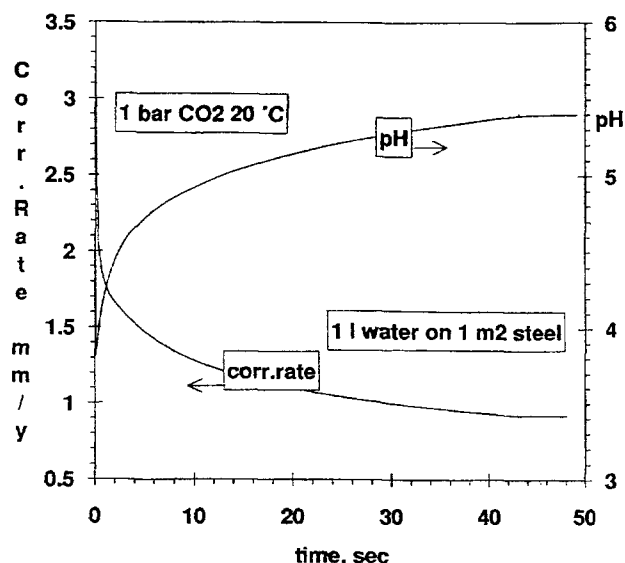


FIGURE 10. Example of calculated pH and corrosion rate as a function of time (conditions as shown).

It should be appreciated however, that Equation (10) was derived for very low CO_2 pressures.

It is recognized that an increase in pH can also lead to the precipitation of protective scales from the water, CaCO_3 , for example. This is not included in the model at present.

INFLUENCE OF FLOW RATE ON Fe^{++} SATURATION

In order to demonstrate the influence of the amount of water in a corroding system, spreadsheet calculations were made by numerical integration of the effect of pH with time. By using straightforward calculations based on balance of ionic charges for a number of small time steps, the pH can be predicted as a function of time up to the point where saturation occurs. An example is given in Figure 10. Depending on the ratio of water volume to surface area and on the severity of the corrosion, the time needed to reach $\text{FeCO}_3/\text{Fe}_3\text{O}_4$ saturation can be significant. This implies that within this time corrosion rates can be experienced that are higher than the nomogram predicts. For the case that Fe-free water enters a pipeline, this means that a certain length will corrode faster than V_{nomo} . In the example of Figure 10, it takes 45 seconds to reach saturation; for a 1-mm water film traveling in a pipeline with a flow rate of 1 m/s, this means that a length of 45 m has higher corrosion rates than predicted with the nomogram. For a completely water-filled pipeline, these lengths would be much larger and dependent on diameter.

Whether or not saturation will be reached also depends on the inflow of "new" water for a given situation.

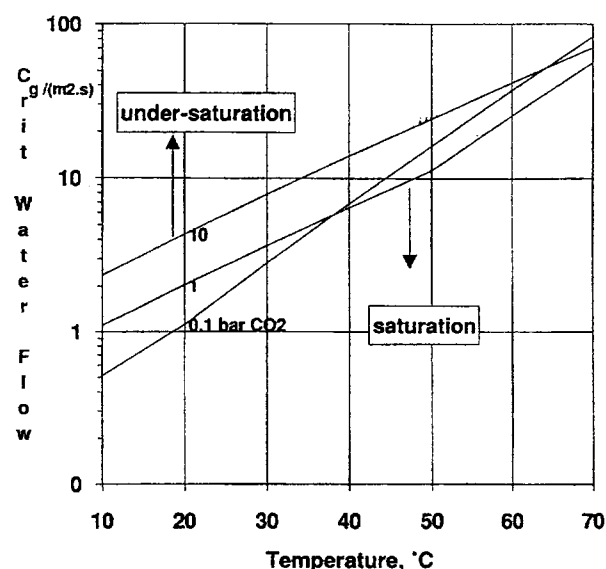


FIGURE 11. Critical inflow of iron-free water below which saturation with $\text{FeCO}_3/\text{Fe}_3\text{O}_4$ is maintained. Higher flow rates lead to undersaturation and corrosion rates higher than predicted with the nomogram.

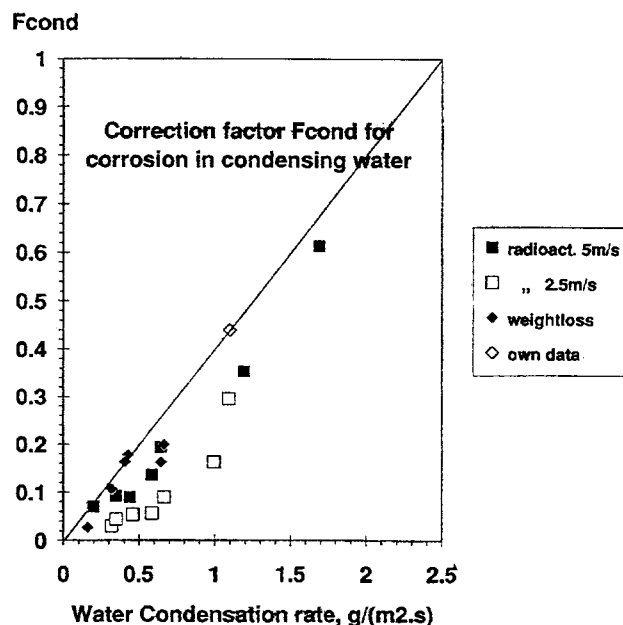


FIGURE 12. Experimental¹⁷ data for condensation factor as a function of condensation rate.

Figure 11 gives the maximum amount of fresh, CO_2 -saturated water that can be accommodated without losing saturation for a given area of corroding steel. This may be referred to as the "critical water inflow." For example, at one bar CO_2 at 50°C , an inflow of more than 10 g of water per second per m^2 would cause undersaturation of FeCO_3 (or Fe_3O_4) and a lower pH, and corrosion rate predictions would have to be corrected using Equation (9) to give a higher rate. A lower influx will maintain saturation. This is especially relevant

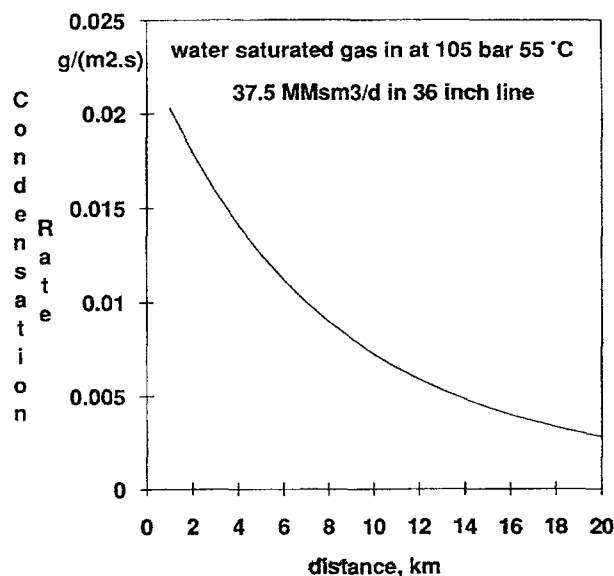


FIGURE 13. Example of water condensation rates calculated for a wet gas pipeline.

for systems with surfaces where water condenses, like in cooler tubes and in slowly cooling wet gas transport lines, as discussed below.

EFFECT OF HYDROCARBON LIQUID

The presence of crude oil, for example, in a line with "live" oil and gas, can have a beneficial effect on corrosion by CO_2 . In the model this is taken into account with the oil factor F_{oil} .

$$F_{\text{oil}} = 0 \text{ if water cut} < 30\% \quad (11)$$

and $V_{\text{crude}} > 1 \text{ m/s}$

otherwise $F_{\text{oil}} = 1$. This expresses the view that the steel can be expected to be oil-wetted (and hydrophobic) if all water is entrained in the crude. If the flow rate V_{crude} of the oil is too low, water can separate and cause corrosion on the bottom of the line. This critical flow rate can be calculated,¹⁴ and is less than 1 m/s for normal crudes. At higher flow rates, the water will be dispersed in the oil. Work by Lotz et al.¹⁵ indicate that in that case at least 30 percent water can be accommodated before the steel is water-wetted. It should be emphasized that light hydrocarbon condensates, for example, natural gas liquids, do not offer any protection in the absence of an inhibitor, regardless of the water content.

CONDENSATION FACTOR

Work by van Bodegom et al.¹⁶ has shown that corrosion rates of steel exposed to condensing water phase in a CO_2 atmosphere quickly decrease with time.

Averaged over time, these rates can be 1/3 to 1/10 of the nomogram corrosion rate, depending on the rate of water condensation. For wet water-saturated gas, some very conclusive tests¹⁷ were carried out by IFE, Norway, using pipe with a radioactive segment in the top. The results are in Figure 12, as the ratio observed/nomogram rates as a function of water condensation rate in $\text{g}/(\text{m}^2\cdot\text{s})$. This ratio has been named the condensation factor F_{cond} . The data can be conservatively described by

$$F_{\text{cond}} = 0.4 \times (\text{condensation rate, g}/[\text{m}^2\cdot\text{s}]), \quad (12)$$

when condensation rate < 2.5

$$F_{\text{cond}} = 1 \text{ when condensation rate} \geq 2.5$$

For wet gas transport, cooling rates and flow rates in pipelines are normally such that condensation rates are far below $0.25 \text{ g}/(\text{m}^2\cdot\text{s})$. An example is given in Figure 13. For such systems, taking $F_{\text{cond}} = 0.1$ will cover all conditions in a conservative manner without having to actually calculate the condensation rates. It is important to note that for these slowly cooling systems, the condensation rates are far below the critical water influx, and the water film will be saturated with corrosion product. For temperatures below the scaling temperature, the corrosion product films formed from saturated FeCO_3 solutions are normally not protective since they are very easily washed away. However, it is postulated that the low liquid flow rates associated with water condensation from slowly cooling gas leave this film intact.

TOP-OF-THE-LINE CORROSION

For wet gas transport lines, water and light hydrocarbons can condense from the gas as the temperature drops with distance. For moderate gas flow velocities of less than 16 to 20 m/s, stratified gas/liquid flow will be the predominant flow pattern. When a corrosion inhibitor solution is injected into such a line, it will not protect against corrosion in the freshly condensing liquid at the top of the line. Actual occurrences of top-of-the-line corrosion have been reported,^{16,19} but here the presence of H_2S may have played an important role.

Without inhibition, the corrosion rate in the top of the line will be lower than that at the bottom, to the extent given by Equation (12), which is very conservatively covered by $F_{\text{cond}} = 0.1$. In cases where this still gives corrosion rates that are too high, the injection of glycol may further reduce this.

EFFECT OF GLYCOL

Glycol is often added to wet gas pipelines to prevent the formation of hydrates. Normally this is mono-

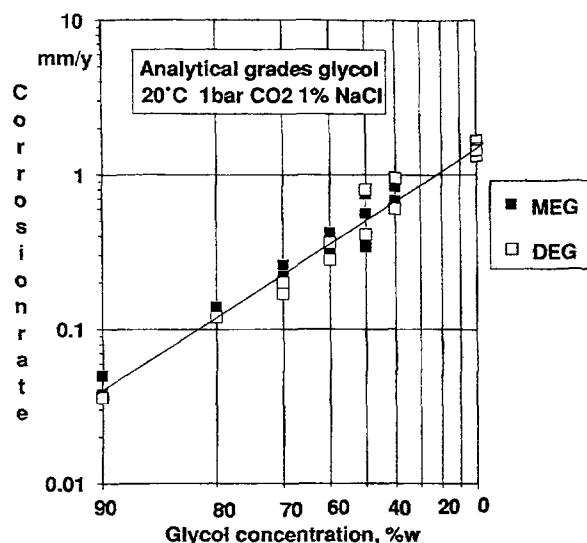


FIGURE 14. Example of effect of glycol on corrosion rate.

or di-ethylene glycol (MEG or DEG), but triethylene-glycol (TEG) is also possible. The presence of glycol also acts on corrosion by CO_2 in two ways:

(1) by reducing the corrosivity of the water phase it mixes with, and

(2) by absorbing water from the gas phase.

The effect of glycol on CO_2 corrosion was systematically screened by Veritec²⁰ tests with tiny corrosion coupons (volume/surface ratio about 10 l/cm²) of two grades of carbon steel in glycol/water mixtures at 20 and 40°C, for various types and grades of MEG and DEG, both with weight loss and polarization resistance measurements. For TEG, time-averaged weight loss data ranging over 90 to 380 h of exposure from a different source were used.²¹ Data for methanol, which can also be used, are not included in the model because of high scatter in the results. For analytical grades of glycol, the effect on corrosion rate could be expressed as a multiplier F_{glyc} :

$$\log F_{\text{glyc}} = A \log (W\%) - 2A \quad (13)$$

W% = water content of water/glycol mixture, in %w

The data showed A to depend only weakly on type of glycol, and A = 1.6 can be used for the model for all glycols. Figure 14 shows an example of the data for 20°C. For various technical grades corrosion rates were less than or equal to those in analytical grade glycols.

The drying action of injected glycol will lower the water dewpoint of the gas. This means that pure water cannot condense as long as the temperature does not fall below this dewpoint. It does not mean that nothing can condense at all: small amounts of glycol/water mixtures may still condense together with hydrocarbon

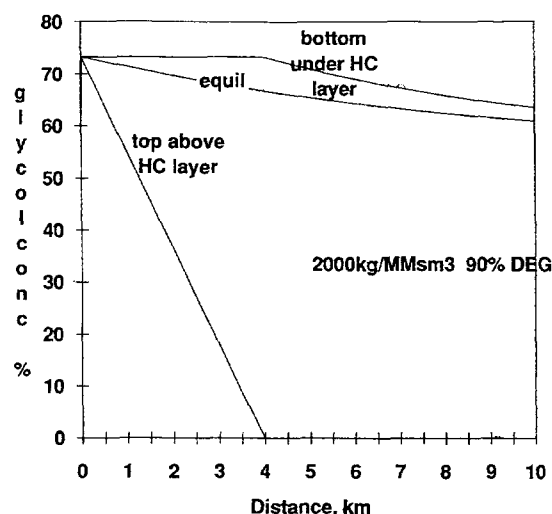


FIGURE 15. Example of distribution of glycol injected in a wet gas pipeline, with and without hydrocarbon layer. Stratified flow (conditions as stated).

liquids. For a system in complete equilibrium, the composition of the condensing glycol/water would be the same as that accumulated on the bottom. Runs with a computer program, which can simulate the effect of deviation from equilibrium in practical cases,²² showed the condensing phase to be about 10 percent leaner in glycol than in the stratified liquid on the bottom for some conditions of practical interest. For prediction of the corrosion in the condensing liquid, which is of little consequence in view of the conservative value used for F_{cond} and can be ignored. Hence the glycol concentrations for the stratified liquid on the bottom and in the condensing phase can be assumed to be equal for practical purposes. The equilibrium composition of glycol/water can be linked to the water dewpoint t_{dew} of the gas at any temperature t by the following formula:

$$t_{\text{dew}}(^{\circ}\text{C}) = \left(-0.038 \frac{115}{115 - \text{DEG}\%} + 1.038 \right) t \quad (14)$$

$$- 1.58 \frac{\text{DEG}\%}{102.2 - \text{DEG}\%}$$

with DEG% = glycol concentration in liquid, in %w. Equation (14), which is for DEG, was derived from a best fit to published data,²³ and holds well for concentrations up to 95%. Similar equations can be derived for MEG and TEG. The dewpoint of the gas is related to the amount of water vapor in the gas at pressure P:

$$\log(\text{water, kg/MMsm}^3) = - \frac{2197.82}{t_{\text{dew}} + 273} \quad (15)$$

$$- 0.89186 \log(P) + 11.67283$$

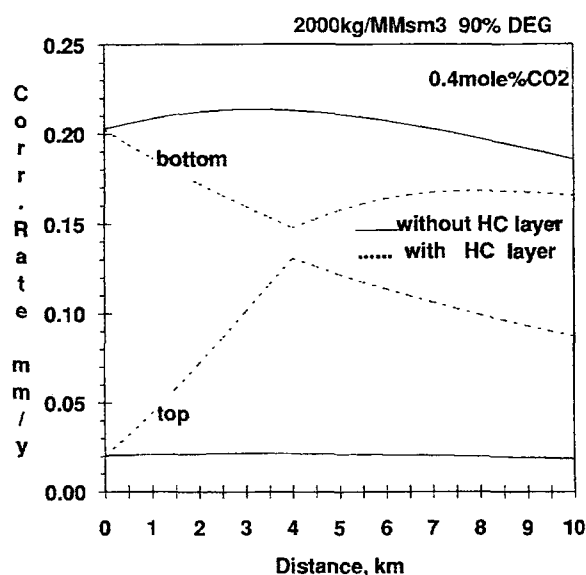


FIGURE 16. Corrosion rates for example in Figure 15.

which describes the graph from McKette and Wehe.²⁴ Equations (14) and (15) can be used to predict the effect of mixing DEG and wet gas. Using the concentration of the injected DEG in Equation (14), the resulting dewpoint can be inserted into Equation (15) to calculate how much water the gas will release by absorption into the glycol. However, this will dilute the glycol, and a number of iterations (normally less than 10) are needed to calculate the equilibrium concentration. With commonly used computer spreadsheets, this approach can be used to calculate the glycol concentrations along the length of a pipeline as a function of gas flow rate and glycol injection rate, together with the resulting corrosion rates, by applying Equation (12). An example of a glycol concentration profile is given in Figure 15 with corresponding corrosion rates in Figure 16. The above calculation is only correct when the contact between the glycol water on the bottom and the gas is not restricted by the presence of a stratified layer of hydrocarbon condensate on top of the glycol. In the presence of such a layer, this cannot always be guaranteed. To predict what the effect would be in such a situation, computer calculations were made for the composition of the condensing liquid for various degrees of contact between gas and glycol. For the worst case of complete blockage, they showed that downstream of the glycol injection point, concentration of the glycol reduced almost linearly to the point where the pipe-wall temperature has cooled below the water dewpoint. Here 100-percent water condenses, and corrosion is only reduced by the condensation factor F_{cond} , and of course, by the lower temperature. This "worst case" scenario can be included in the model calculations described above.

The possible effect of a hydrocarbon (HC) layer is included in Figures 15 and 16. The likelihood of this situation is probably low since it would imply that gas hydrates could form in the top of the pipe, which is not a recognized problem in this situation. The presence of a stratified-wavy flow pattern is likely to reduce this risk even further.

INHIBITION

The effect of an inhibitor can be included in the model by simply dividing corrosion rates by an inhibitor efficiency factor, for example, for an inhibitor with 90 percent efficiency, the corrosion rate should be divided by 10. For stratified flow patterns, the model will automatically account for the likely absence of inhibitor in the top of the line. Furthermore, Geelen and Groenewoud reported that gas velocities exceeding 17 m/s may affect inhibitor performance.²⁵

DISCUSSION

The model provides correction factors to be applied to the nomogram corrosion rate V_{nomo} (Equation [1]) to obtain a conservative estimate of CO_2 corrosion rates. As a first order approximation, correction of V_{nomo} is to be considered for each factor that differs from 1. In some cases, this could lead to application of more than one factor, all significantly differing from 1, which could influence each other, or in the worst case, might be mutually exclusive. The combination of the scale factor F_{scale} with the condensation factor F_{cond} falls in this category, since they both relate to protection from a corrosion product film. Starting with a low-temperature film, an increase in temperature will change the film's texture to a more adherent one. Since no quantitative information is available at present about this, the following conservative point of view is incorporated in the model: when the scaling temperature is exceeded, the corrosion rate is kept constant at the value of $V_{nomo} \times F_{cond}$ at $t=t_{scale}$, until the temperature is reached where this rate is larger than $V_{nomo} \times F_{scale}$. Although it is probable that the corrosion rate decreases when the scale temperature is exceeded, no further reduction of the corrosion risk will be predicted in the model until the high-temperature scale alone will lead to a lower rate. At present, the combination of F_{scale} with F_{pH} (to account for bulk pH) is not allowed in the model; the corrosion rate of scale-covered steel is more likely to be controlled by the pH and Fe^{++} concentration resulting from local saturation with $FeCO_3$ or Fe_3O_4 at the steel's surface.³ For this reason, F_{pH} is set = 1 when $F_{scale} < 1$.

Application of F_{glyc} and F_{cond} at the same time is allowed since the steel can be expected to "see" the glycol in the presence of a low-temperature $FeCO_3/Fe_3O_4$ film. Combination of F_{glyc} and F_{pH} is allowed be-

cause different experiments on the effect of glycol with different surface area/volume ratios (leading to different pH values) give very similar results for F_{glyc} , suggesting that these effects are independent. Similarly, at high temperatures, the remaining corrosion rate of scale-covered steel in the alkaline micro-environment at the steels's surface is expected to be affected by the presence of glycol approximately in the same manner, which implies that multiplication with both F_{glyc} and F_{scale} is allowed.

With these considerations, the equations for the various factors can be combined to predict corrosion rates in wet gas lines. They can be incorporated in any computer program, although in practice it appears that personal computer-based spreadsheets form an ideal vehicle for these calculations because they are inherently transparent to the user.

CONCLUSIONS

- ❖ The corrosion rate predicted by the de Waard-Milliams equation for CO_2 corrosion holds for water that is saturated with corrosion product. A correction can be calculated in case the water is undersaturated.
- ❖ The effect of dissolved Fe^{++} on CO_2 corrosion rate can be accounted for through the effect of changes in pH caused by the Fe^{++} . Using a theoretically derived pH shift at saturation, the observed difference between "once-through" and constant volume corrosion tests can be reproduced over a temperature range of 20 to 80°C.
- ❖ It is possible to calculate the flow of water needed to maintain undersaturation with FeCO_3 or Fe_3O_4 . In practical cases, the flow of water condensing in a pipeline system can be expected to be less than this, that is, saturation will occur, and the deWaard-Milliams equation can be applied.
- ❖ The temperature where the de Waard-Milliams equation is applicable can be extended to higher temperatures by using a correction factor to account for the protection by corrosion product films formed at these temperatures. This factor implies the existence of a temperature where the corrosion rate goes through a maximum, which is called the scale temperature.
- ❖ At lower temperatures, the film formed in condensing water in the top of a pipeline gives significant protection because of the low flow rates involved here.
- ❖ The reduction of corrosion rates to be expected when glycol is injected into a wet gas transport line can be calculated along the length of the line with the proposed model.

❖ Combination of the correction factors used to account for various effects is possible in some, but not all, instances.

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